

Based on the document provided, here are the extracted CLI commands for the Router (R1) and the PCs. I have cleaned up the typos and organized them into executable steps.

Step 1: User Access Authentication (R1)

Cisco CLI

```
enable
configure terminal
username admin secret cisco
aaa new-model
aaa authentication login default local
line console 0
  login authentication default
exit
exit
```

(Note: I replaced `xxxx` and `yyyy` with `admin` and `cisco` for the username and secret, as these are required for the `aaa` configuration to work properly).

Step 2: Enable Security License (R1)

Cisco CLI

```
enable
show version
configure terminal
license boot module c1900 technology-package securityk9
yes
end
copy running-config startup-config
reload
```

(After the router reloads, you must log in again)

Cisco CLI

```
enable
show version
```

Step 3 & 4: Connectivity Check (PC Command Prompt)

PC0:

DOS

ping <IP_Address_of_PC1>

PC1:

DOS

ping <IP_Address_of_PC0>

Step 5: Configure IPS (R1)

Create Directory and Basic IPS Config

Cisco CLI

```
enable
mkdir ipsdir
! Press Enter to confirm the folder name "ipsdir"
configure terminal
ip ips config location flash:ipsdir
ip ips name iosips
ip ips notify log
exit
```

Set Clock and Logging (Privileged EXEC & Global Config)

Cisco CLI

```
clock set 19:25:59 9 July 2023
configure terminal
service timestamps log datetime msec
logging host 192.168.1.50
```

Configure Signature Categories

Cisco CLI

```
ip ips signature-category
category all
retired true
exit
```

```
category ios_ips basic
retired false
exit
exit
! Press Enter to confirm changes
```

Apply to Interface

```
Cisco CLI

interface g0/1
ip ips iosips out
exit
```

Modify Signature (Signature 2004: ICMP Echo Request)

```
Cisco CLI

ip ips signature-definition
signature 2004 0
status
retired false
enabled true
exit
engine
event-action produce-alert
event-action deny-packet-inline
exit
exit
exit
! Press Enter to confirm changes
end
```

Verify Configuration

```
Cisco CLI

show ip ips all
```

Step 6 & 7: Testing (PC Command Prompt)

Ping PC1 (Should Fail/Timeout):

DOS

ping <IP_Address_of_PC1>

Ping PC0 (Should Success):

DOS

ping <IP_Address_of_PC0>